THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MECHANICAL ENGINEERING

IDAT7100 Capstone Experience

Interim Report

2025-2026

Project Title:	Boundary Evaluation of Low-Fidelity sEMG Acquisition for Dexterous Hand Control
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Date:	01/04/2026

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Introduction

1.1 Research Background

In recent years, with the development of robotic and artificial intellectual technologies, the development of dexterous hands has transitioned from traditional rigid industrial grippers to highly biomimetic, high-degree-of-freedom (DOF) systems. Dexterous hands are advanced robotic end-effectors designed to mimic the anatomical structure and kinematics of the human hand, possessing the potential to perform complex, human-like manipulations [1], [2]. To achieve intuitive and seamless control over these mechanical structures, surface electromyography (sEMG) has emerged as the most prominent non-invasive Human-Machine Interface (HMI). By capturing electrical activities generated during muscle contractions, sEMG directly decodes human motor intentions.

Within this sEMG field, there are already commercialised products, and for which are at the top of this field are the company like BrainCo. For BrainCo, they are known for the technologies related to non-invasive brain-computer interfaces (BCI) and surface electromyography (sEMG) control. Based on their latest model of dexterous hand, Revo 2, it has reached a new level of total weight 383 g and highly compact design, and can still produce an astonishing gripping force of 50 N and maximum lay load capacity of 20 kg, which is a very high grip force-to-weight ratio [3].

To achieve such performance, commercial leaders typically rely on high-density sEMG sensor arrays for low-latency and high-accuracy intention recognition. While in academic research, to process the sEMG signals properly in real world, the properties of inherent noise and non-stationary must be conquered. Most advanced studies have applied advanced noise reduction techniques, like Ensemble Empirical Mode Decomposition (EEMD), to mitigate motion artifacts and environmental interference. Following signal conditioning, non-linear time-series features are extracted using methods like the Multivariate Autoregressive (MVAR) model [4]. For such high-computing-power architectures, making high-level of hand operation possible and study has shown that optimised real-time classification models under the laboratory condition have the ability to decode up to 13 distinct hand gestures with a prediction efficiency of 92.7% [5]. All of these represent the highest level of intent-decoding technologies nowadays.

1.2 Problem Statement

Despite of the excellent performance achieved by commercial bionic hands and advanced laboratory models, their application is still severely restricted by high costs and strict hardware requirements. These systems highly rely on high-density sEMG sensors or arrays, unique microprocessors and powerful computing capabilities. Consequently, this "Hardware-stacking" paradigm not only making it economically difficult achieve wide usage of these products but also significantly arises the barriers to entry for open-source foundational research prototype development [6].

Furthermore, achieving reliable and intuitive control over myoelectric bionic hand is fundamentally difficult due to the inherent non-stationary nature of human sEMG signals. Research indicates that many potential users, particularly amputees, struggle to master traditional myoelectric control or experience declining performance over time, as they must continuously adapt to develop alternative muscle synergy patterns [7]. In traditional systems, this adaptation imposes a heavy cognitive burden on users, requiring extensive and tiring training, and users rely on visual feedback to consciously enforce different muscle activations [7]. To shift this learning burden from the human user to the machine, advanced supervised learning models are employed. However, training these models to automatically map unstable sEMG signals requires highly precise physical hand kinematics as the objective "ground truth." Traditionally, acquiring such synchronized ground truth relies on prohibitively expensive optical motion capture systems or sophisticated data gloves, making the difficulty of multimodal data collection for low-cost systems more difficult [8].

To overcome these financial and technical constraints, utilizing commercially available off-the-shelf bioelectric analogue front-ends (such as AD8232, originally designed for ECG acquisition) provides a feasible low-cost alternative [9]. However, a critical physical conflict is that: the AD 8232 hardware filtering characteristics of these low-cost modules inevitably eliminate the high-frequency discharge details of muscle fibres, primarily retaining the low-frequency signal envelope. Traditional complex regression algorithms, designed for independent, multi-finger continuous control, typically require broadband, high-fidelity sEMG signals and fail when applied to such narrowband signals [10]. Therefore, the problem would be how to achieve smooth and compliant proportional grasping control without modifying the underlying low-cost hardware circuitry—rather than rigid, discrete gesture classification.

1.3 Research Objectives

The main objective of this study is to evaluate the performance limits of myoelectric dexterous hand control under strict hardware constraints. To achieve this goal, this project aims to accomplish the following specific objectives:

- **Hardware Assembly and Debugging:** Assemble and debug the existing open-source TetherIA mechanical hand, which serves as the physical platform for this project.
- **Signal Acquisition System Development:** Build a complete signal acquisition system from scratch. This includes generating the Bill of Materials (BOM), purchasing necessary modules, and writing the related code for the microcontroller.
- **Algorithm Evaluation Framework:** Establish a systematic framework to assess and compare the accuracy of various machine learning algorithms when processing signals from low-density, consumer-grade surface electromyography (sEMG) sensors.
- **Progressive Validation Pipeline:** Implement a phased testing approach. Once an algorithm achieves satisfactory accuracy, validation will advance to simulation-to-

simulation testing, followed by simulation-to-reality deployment.

The main contribution of this project is the provision of a structured evaluation pipeline that identifies how far low-cost hardware can be pushed through algorithmic compensation. This framework aims to offer a clear, systematic reference for future accessible bionic research.

2. Methodology

This section details the methodology employed to evaluate the myoelectric control framework. The overall system architecture is illustrated in Figure 1. The process is divided into several different stages, first collecting sEMG signals, then digital preprocessing, while performing real labelling, merging these labels, conducting algorithm evaluation, and finally physical driving of the TetherIA hand.

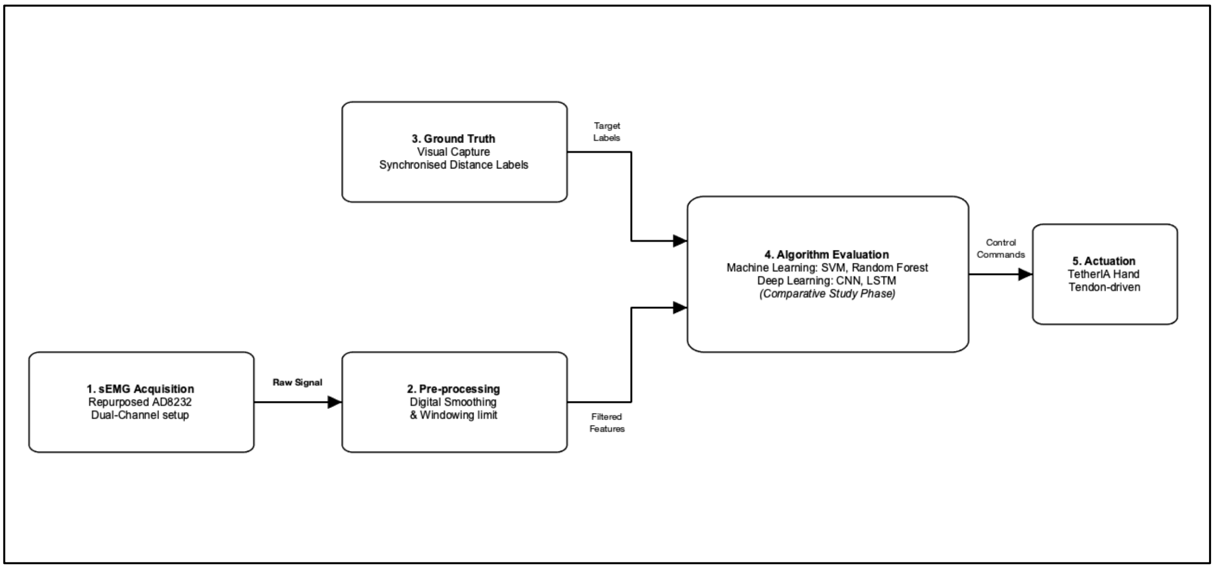

Figure 1, Block diagram of the system architecture, illustrating the data flow from sEMG signal acquisition to the final actuation of the mechanical hand.

2.1 Hardware Platform and Signal Acquisition:

The physical actuation platform for this project is based on the TetherIA hand, an open-source, tendon-driven dexterous hand. The work on this is focussed on the assembly and debugging of this hand, and the assembly is followed by assembly instruction provided by TetherIA while the Bill of Material is modified due to reasons like: some of the components i.e., some springs, are non-standard components, therefore, BOM need to change to mitigate the situation. Also, this BOM is for the entire project, therefore, it also includes the materials for other modules like sEMG modules setup.

Following the assembly of TetherIA hand, a custom signal acquisition system has been built from scratch to capture human intent. This system utilises a low-density setup with two consumer-grade surface electromyography (sEMG) sensors. These sensors are placed on specific muscle group to detect electrical signals. After the muscle analogue signals received by sensors, they will then be processed by an ESP32 (marked as ESP-32D) microcontroller, which handles the analogue-to-digital (A/D) conversion and data transmission. This low-cost, dual-channel setup serves as the physical baseline for the algorithm evaluation framework.

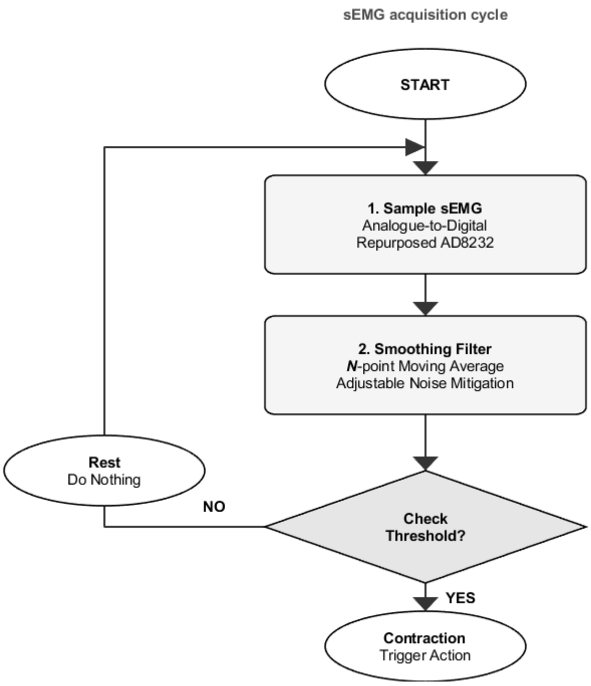

Figure 2, Flowchart of the sEMG acquisition cycle, illustrating the logic from signal sampling to contraction detection.

2.2 Data Pre-processing:

The raw signals captured by the ESP32D microcontroller are often subject to electrical noise and physiological fluctuations due to the use of consumer-grade sensors [11]. To ensure the data is suitable for algorithmic analysis, a two-stage preprocessing sequence is implemented:

First is Digital smoothing: A digital filter, specifically a moving average, is applied to the raw sEMG data. This step aims to reduce high-frequency noise and calibrate the signal, providing a smoother envelope that more accurately represents muscle intensity.

Next is Windowing: The continuous data stream is segmented into discrete time intervals, or windows [12]. This is essential for real-time applications, as it defines the amount of data processed in each iteration. By segmenting the signal into manageable discrete intervals, the system can provide a steady flow of filtered features to the subsequent classification models [13].

The output of this stage consists of structured, filtered features, which serve as the direct input for the algorithm evaluation framework.

2.3 Ground Truth Synchronisation:

To evaluate machine learning algorithms, a dataset containing precise motion labels is needed. Additionally, a custom synchronisation program is required to match physical surface electromyography signals with visual motion labels.

As illustrated in Figure 3, the data collection pipeline follows a clear progression. First, a standard webcam is used for video input, in this case would be the camera on the laptop. The MediaPipe framework is then employed for landmark detection, identifying the right hand and tracking specific anatomical joints [14]. For feature extraction, the spatial relationship between the fingers is mathematically quantified. Let the coordinates of the thumb tip be $P_0(x_0, y_0)$, and the coordinates of the rest four fingertips (index, middle, ring, and Pinky) be $P_i(x_i, y_i)$, where $i \in \{1,2,3,4\}$. The 2D Euclidean distance d_i between the thumb and each respective finger is calculated as:

$$d_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2} \tag{2.3.1}$$

This calculation yields a four-dimensional visual feature vector, $V = \{d_1, d_2, d_3, d_4\}$, representing the physical closure state of the hand at any given timestamp [15].

To pair the muscular intent with the physical execution, the sEMG values received via sensors are continuous, with this visual feature vector. A unified timestamp is recorded for each data frame, ensuring that the continuous sEMG data and corresponding hand coordinates are strictly aligned in time. Precise time synchronisation of multimodal data is a fundamental requirement for ensuring data integrity in biomechanical pattern recognition [16].

Ultimately, this synchronised logging process outputs a structured dataset. This dataset directly serves as the labelled ground truth required for training, validating, and evaluating the predictive models in the subsequent machine learning framework [17].

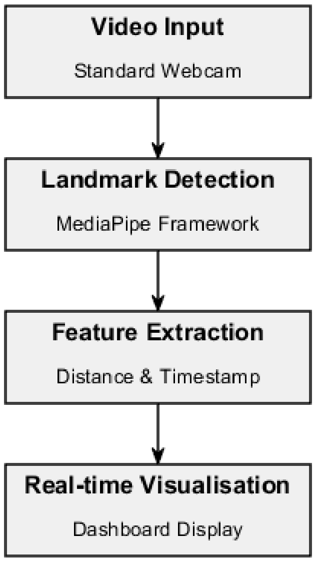

Figure 3, Flowchart of the ground truth synchronisation process, detailing the extraction of visual features using the MediaPipe framework.

2.4 Algorithm Evaluation Framework

The final stage of the methodology established a systematic framework for evaluating the predictive performance of different algorithmic methods. By utilizing a synchronised sEMG-visual dataset, the system compared four different models to determine the optimal balance between classification accuracy and computational efficiency in real-time bionic control.

As shown in the Figure 1, the selected algorithms are categorised into two sets. To structure this selection, these algorithms are listed into four different levels based on their complexity. (as shown in Table 1):

Level 1 & 2 Traditional Machine Learning: Support Vector Machines (SVM) serve for Level 1, targeting basic binary recognition (e.g., rest versus action). For Level 2, Random Forest (RF) is implemented to handle complex binary or basic multi-class classification.

These models were chosen because they have established robustness in high-dimensional feature spaces and have relatively low computational overhead [18].

Level 3 & 4 Deep Learning: To achieve more complicated operation control, non-linear relationships and temporal dependencies within the sEMG signals, Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks are employed, also as State-Of-The-Art models. Level 4 is set as a stress test. The goal of this final phase is not to achieve state-of-the-art accuracy, but rather to find the exact mathematical boundary where the physical hardware bottleneck can no longer be compensated by advanced deep learning algorithms

Level 1: Baseline Binary Classification	Level 2: Advanced State Classification	Level 3: Temporal Sequence Analysis	Level 4: Complex Intent Recognition
Support Vector Machine (SVM)	Random Forest	Long Short-Term Memory (LSTM) network	CNN
Basic Binary recognition (rest vs action	Complex binary or basic multi-class recognition with increased gesture variations	Continuous gesture recognition leveraging long-term temporal dependency analysis	Fine-grained muscle intent recognition and force/angle prediction for natural and precise control

Table 1, Four-tier algorithm evaluation framework. The strategy demonstrates the progression from basic binary rest-action classification using traditional algorithms to fine-grained intent and force prediction using advanced neural networks

Before the model training, the structured dataset needs feature scaling (e.g., standardisation). This step normalises the numerical ranges for both the sEMG and visual data, make sure that features with naturally large magnitudes do not bias the learning process of the algorithms. After that, the scaled data is processed using a k-fold cross validation strategy. This iterative training and testing approach guarantees that every data point is utilised for both phases, therefore, preventing overfitting [19]. Finally, after all the models are trained with the dataset, comparison will be conducted using a confusion matrix. Based on the performance of confusion matrix, evaluations can be performed like F1-Score.

3. Preliminary Results and Discussion

So far, the assembly of TetherIA right hand is finished. During the assembly there were multiple problems found, which would be summarised in the following:

When purchasing all the components, i.e., springs, pins, everything is followed by the BOM provided by TetherIA although some components need used substitutes as they are non-standard components. May due to the keep updating of the repository, the BOM cannot perfectly match with the assembly instructions provided, also due to lack of robotic assembly experience, the assembly process will inevitably be time-consuming and may result in damage to components.

Despite of that, the TetherIA hand was assembled completely and microcontroller was programmed, which made it possible to control and tuning the servos inside the robotic arm later.

In parallel with the TetherIA hand assembly, the data acquisition hardware was purchased and assembled. The circuit is currently mounted on a breadboard, integrating an ESP-32D microprocessor with an AD8232 biopotential module adapted for collecting sEMG signals. The microprocessor was programmed to manage the continuous acquisition and analogue-to-digital (A/D) conversion of the raw bioelectric signals. Initial feasibility tests confirmed the successful capture of these signals, showing that the hardware-to-software data pipeline is fully operational and prepared for subsequent algorithmic processing.

Furthermore, a vision-based framework utilising MediaPipe was successfully deployed to track hand kinematics in real time. This setup extracts spatial features, making ground truth labelling required for the machine learnings.

Figure 4, Damage during assembly

Figure 5, Demonstration of all the components purchased and printed components

Figure 6, Assembled hand front view operated via GUI

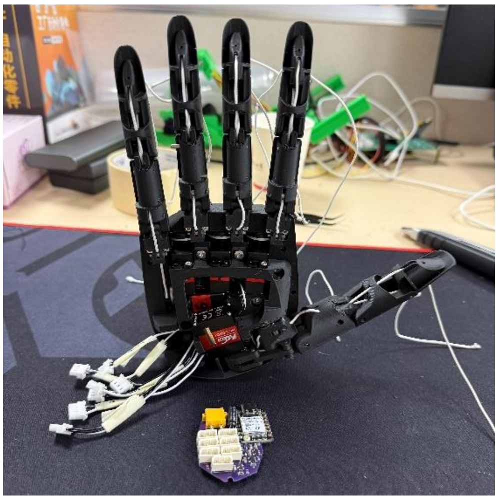

Figure 7,Demonstration of servo motors and microcontroller at the palm

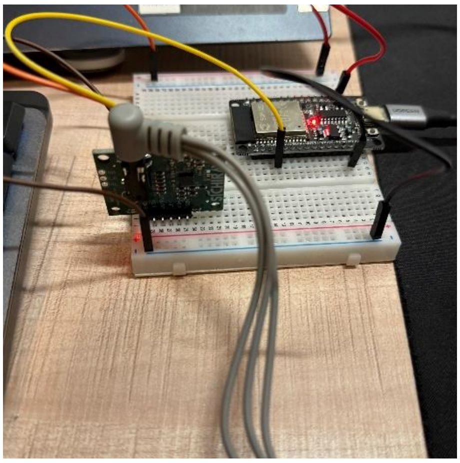

Figure 8, Demonstration of sEMG sensor module

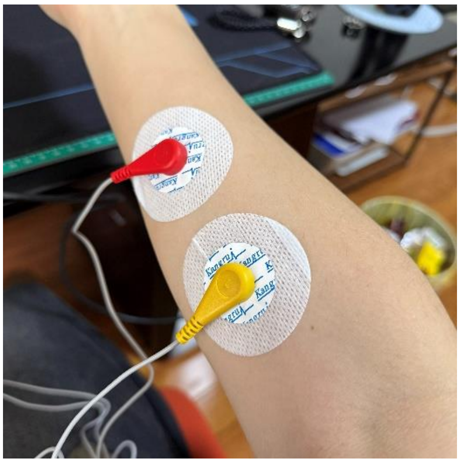

Figure 9, Demonstration of sEMG sensors on forearm

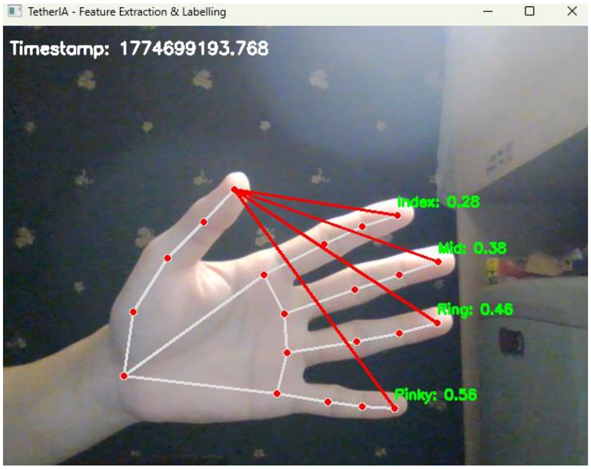

Figure 10, Real-time hand landmark tracking implemented via the MediaPipe framework, capturing spatial features to serve as kinematic ground truth for future sEMG labelling.

4. Conclusions and Future Work

So far, the hardware and software required for bioelectric acquisition and dataset labelling are successfully established. Despite minor challenges during the assembly and testing, the assembly of TetherIA hand has been assembled and tuned well.

For future work, the project will change into the dataset acquisition and processing, algorithm evaluation, and deployment phases. The tasks are structured according to the defined work packages in Table 2 under Appendix.

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Appendices

Work Package	Task ID	Task Description	Prerequisites
WP1: Data Pipeline & Hardware Setup	T1	Procurement and integration of 2nd sEMG Module	None
	T2	Hardware Debugging & Signal Pre-processing Optimisation	T1
	T3	Development of Automated Data Annotation Pipeline via visual synchronisation	T2
WP2: Progressive Algorithm Evaluation	T4	Level 1& 2 tuning: Baseline binary and advanced state classification (SVM, Random Forest)	T3
	T5	Level 3 tuning: Temporal sequence analysis for continuous gesture recognition (LSTM)	T4
	T6	Level 4 tuning: Complex intent recognition and fine-grained prediction (CNN)	T5
WP3: System Validation & Deployment	T7	Kinematic simulation validation (Sim-to-Sim)	T6
	T8	Physical deployment and tendon-driven actuation testing (Sim-to-Real)	T7
WP4: Report Writing	T9	Outline finalisation, Introduction and literature review	None
	T10	Methodology drafting	T3
	T11	Result analysis, Discussion and Future work	T8
	T12	Final report content and formatting check	T11

Table 2, Future Plan

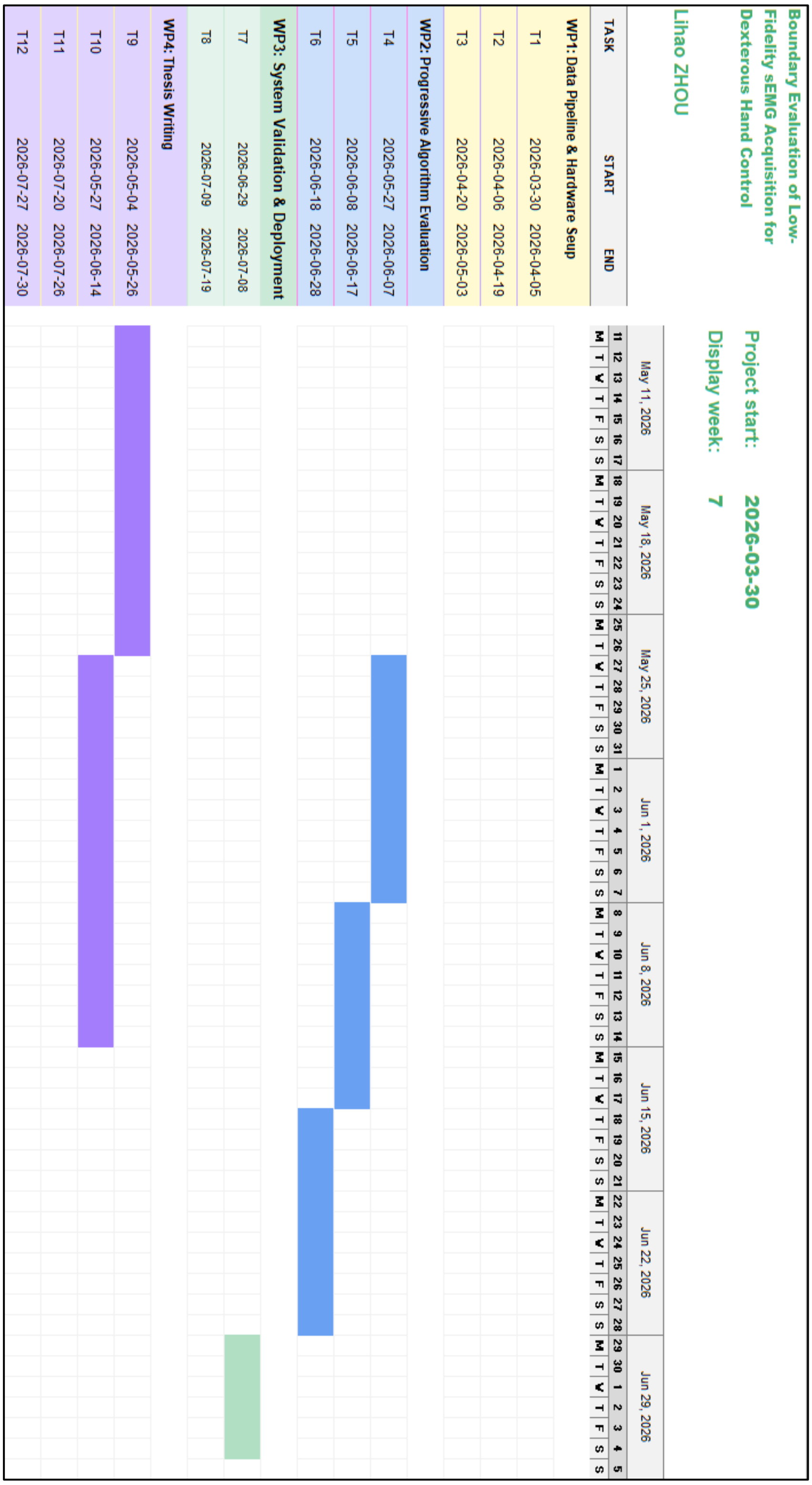

Figure 11, Gantt Chart demonstration for future plan